

the probability of success of the treatment at one versus the other, the patient’s willingness to tolerate invasive operations, the importance of being close to relatives and friends, and a host of other reasons. Consequently, there is no doubt that given the observed covariates and the home hospital type, the assignment of treating hospital type is confounded. Therefore, doing a direct analysis of treating hospital type, even if propensity score methods were used to create subclasses of patients with identical distributions of all observed covariates in large and small treating hospitals, would be considered unsatisfactory because key covariates were not available in the data set.

We can, however, still make progress based on the assumed unconfounded assignment of home hospital type by using a different template for our observational study of treating hospital type: a randomized experiment with noncompliance. That is, think of patients who transfer, or, more generally, who would have transferred if assigned to a different hospital type, as being noncompliers, and therefore, our template is that of a randomized encouragement design, where the encouragement to be treated in the diagnosing large or small hospital is randomly assigned within propensity score strata. The crucial idea here is then to stratify also on the bivariate “intermediate outcome,” treating hospital when assigned to a large home hospital and treating hospital when assigned to a small home hospital. Even though only one of these intermediate variables is actually observed, progress can still be made. Notice that the design phase does here look at intermediate outcome

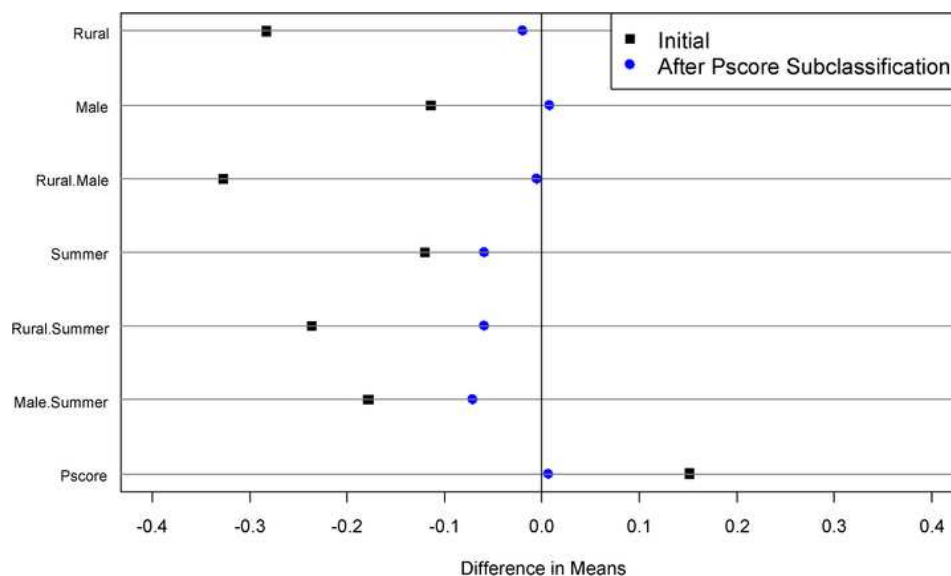


FIG. 7. *Cardia cancer, difference in means for binary covariates and p-score.*

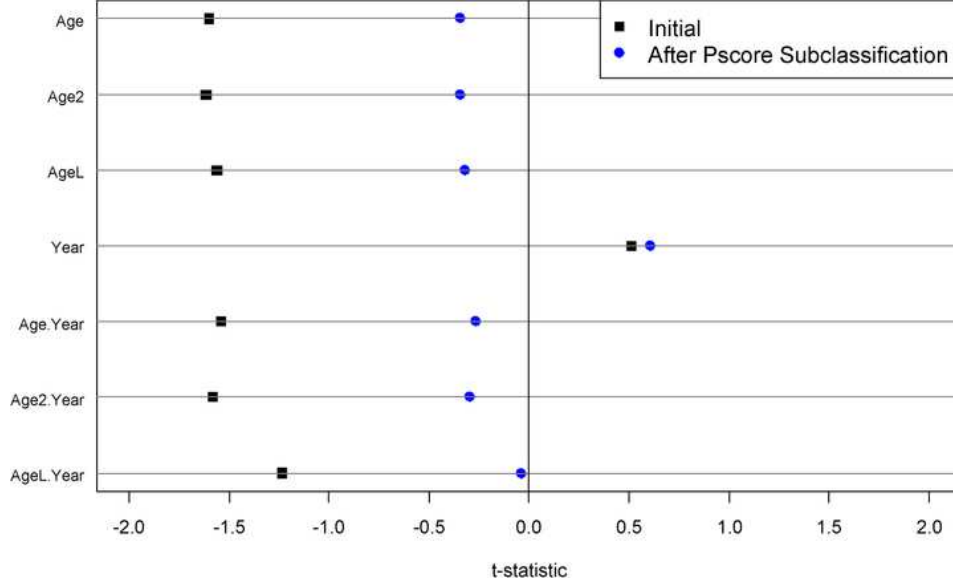
FIG. 8. *Cardia cancer, t-statistics for continuous covariates.*

TABLE 3

Cardia cancer: observed counts in observed groups and approximate counts in principal strata under monotonicity assumption—subclass 1

(1)		(2)	(3)	(4)		(5)	(6)
“Assigned”/ randomized home hospital type		Treating hospital type T	#	Underlying principal strata: $h =$		Approximate proportion in population in principal strata	Approximate N in LS principal stratum
h	#			ℓ	s		
(1)	ℓ	L	5	L	L	44%	3
				L	S	56%	
(2)		S	0	S	S	0%	
(3)	s	L	11	L	L	44%	14
				S	S	0%	
(4)		S	14	L	S	56%	

data, treating hospital type, but not the outcome data on survival, on which decisions will be based. Survival data are not available at this stage!

Denote the home hospital type by h , which takes the value ℓ when assigned large hospital type and s when assigned small home hospital type. Similarly, let T denote treating hospital type, which takes the value L when the treating hospital is large, and takes the value S when treating hospital is

small. The first three columns of Tables 3–7 summarize the observed values of h and T within each of the five propensity score subclasses. Clearly, in all subclasses, transfers into large hospitals are common, but only in subclass 5 are there any $\ell \rightarrow S$ transfers. But do we estimate that there are compliers, who are treated in both large and small treating hospital types, within each subclass? If not, we will not be able to estimate the causal effect of treating hospital type for the entire group of patients—a critical design issue with this template.

TABLE 4
Cardia cancer: observed counts in observed groups and approximate counts in principal strata under monotonicity assumption—subclass 2

(1)		(2)	(3)	(4)		(5)	(6)
“Assigned” / randomized home hospital type		Treating hospital type T	#	Underlying principal strata: $h =$		Approximate proportion in population in principal strata	Approximate N in LS principal stratum
h	#			ℓ	s		
(1)	ℓ	12	L	12	L L	71%	3
(2)			S	0	L S	29%	
(3)	s	17	L	12	S L	71%	5
(4)			S	5	S S	0%	
						29%	

TABLE 5
Cardia cancer: observed counts in observed groups and approximate counts in principal strata under monotonicity assumption—subclass 3

(1)		(2)	(3)	(4)		(5)	(6)	
“Assigned” / randomized home hospital type			Treating hospital type T	#	Underlying principal strata: $h =$		Approximate proportion in population in principal strata	Approximate N in LS principal stratum
h	#	ℓ			s			
(1)	ℓ	17	L	17	L	L	38%	11
(2)			S	0	S	S	62%	
(3)	s	13	L	5	L	L	38%	8
(4)			S	8	S	S	62%	

5.4. *Principal strata and the monotonicity assumption.* Formally in the RCM, there are two types of outcomes: (1) the treating hospital type, T , which equals $T(\ell)$ when $h = \ell$ and $T(s)$ when $h = s$, and (2) the survival time since diagnosis, Y , which equals $Y(\ell)$ when $h = \ell$ and $Y(s)$ when $h = s$. The possible values of $(T(\ell), T(s))$ will be denoted LL, LS, SL , or SS [where, for simplicity, LL means the same as (L, L) , etc.], and those values define four possible “principal strata.” LS can be thought of as the stratum of compliers, that is, nontransfer patients; the LL and SS strata can be thought of as noncompliers who will always be treated at the same hospital type no

TABLE 6
Cardia cancer: observed counts in observed groups and approximate counts in principal strata under monotonicity assumption—subclass 4

(1)		(2)	(3)	(4)		(5)	(6)
“Assigned”/ randomized home hospital type		Treating hospital type T	#	Underlying principal strata: $h =$		Approximate proportion in population in principal strata	Approximate N in LS principal stratum
h	#			ℓ	s		
(1)	ℓ	19	L	19	L	55%	9
(2)			S	0	S	45%	
(3)	s	11	L	6	L	55%	5
(4)			S	5	S	45%	

TABLE 7
Cardia cancer: observed counts in observed groups and approximate counts in principal strata under monotonicity assumption—subclass 5

(1)		(2)	(3)	(4)		(5)	(6)
“Assigned”/ randomized home hospital type		Treating hospital type T	#	Underlying principal strata: $h =$		Approximate proportion in population in principal strata	Approximate N in LS principal stratum
h	#			ℓ	s		
(1)	ℓ	20	L	18	L	67%	5
(2)			S	2	S	23%	
(3)	s	9	L	6	L	10%	2
(4)			S	3	S	67%	